

Matter-first automated discovery of transient spacetime shortcuts

Nikita M. Shirokov*

Independent Researcher, Moscow, Russia

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We present an automated matter-first pipeline for discovering warp precursors—transient spacetime shortcuts. An optimizer proposes scalar matter, **GRTresna** solves the constraints, **GRTeclyn** evolves the full 3+1 Einstein equations, and null geodesics integrated through the *time-evolving* metric test whether light arrives earlier than in flat space. Quality-diversity search over a 39-dimensional bicomplex Q-ball genome yields a five-lump champion (candidate 146) with two canonical and three phantom (negative-energy) sources. Freely falling observers receive the photon 7.6% early by their own clocks—the physical magnitude of the shortcut, against 20.2% between fixed coordinate endpoints—stable across resolution and domain ladders and reproduced with the ignition controller off. Field-by-field ablation attributes the transport to conformal spatial contraction and lapse elevation, not to frame dragging: deleting the shift entirely *increases* the shortcut. A companion pure-geometry atlas explains why. Its kinematic ceiling is 48.5% for shift-dominated (Alcubierre) topologies but only 21.8% for the curvature-dominated branch the champion sources and nearly saturates—a Pareto frontier between kinematic strength and dynamical sourceability. A matched canonical-only search finds no advantage at all. The channel is transient: the matter disperses and the residue collapses to an apparent horizon by $t \approx 27$.

I. INTRODUCTION

The goal of this work is an *automatic discovery pipeline* for faster-than-light (FTL) warp precursors, in the spirit of recent automated theory- and pipeline-discovery efforts in fundamental physics [1, 2]. We define a warp precursor—a transient *spacetime shortcut*—as a constraint-satisfying matter configuration that generates an effective null timing advantage prior to its own dispersal: null rays between prescribed fixed endpoints in the weak-field region traverse the curved spacetime in less coordinate time than the matched flat-space path—never local superluminal motion outside the light cone. “FTL” and “null shortcut” are used interchangeably for that operational claim throughout. Rather than handcrafting a metric and hoping the implied stress-energy is physical, we want a closed loop that proposes candidates, evolves them under the Einstein equations, and retains only those that pass a gauge-invariant transport test.

The standard route is *geometry-first* (equivalently metric-first): one prescribes $g_{\mu\nu}$, reads off $T_{\mu\nu} = G_{\mu\nu}/8\pi$, and studies frozen slices [3–5]. That recipe is poorly conditioned for automated search. Most deformations of $g_{\mu\nu}$ violate the constraints, have no local matter realization, or fail as soon as time advances, so an archive of hundreds of such candidates fills with numerical failures rather than competing mechanisms.

We therefore invert the loop (Fig. 1): the optimizer proposes matter, **GRTresna** builds the constraint-satisfying initial slice, **GRTeclyn** [6, 7] advances the spacetime, and probes ask whether a null shortcut appears (the two codes are detailed in Sec. II A). The fractional advantage is f_{geo} [Eq. (13)], quoted throughout as a percentage: $f_{\text{geo}} = 20\%$ means the curved transit is one fifth shorter

than the flat comparison. By classical theorems on superluminal travel and gravitational time delay [8–10], any such advantage between endpoints in the (near-)flat region requires violation of the null energy condition, and generic warp-drive spacetimes share this requirement [11]; the phantom sector introduced below is the controlled realization of that necessity in our matter model.

This criterion is the acceptance test, not a post-hoc diagnostic. A warp precursor is useful only if it changes what light can do between fixed distant observers, and local coordinate signal speeds, lapse and shift artifacts, or a positive frozen-slice diagnostic can all look “superluminal” while no ray actually connects the endpoints sooner once the metric evolves.

Equally important is the *scoring metric* that turns each evolution into the fitness $S(\theta)$ [Eq. (9)], because MAP-Elites converges to whatever S rewards. Earlier campaigns with weaker objectives reliably “cheated”—harvesting coordinate proxies while the matter dispersed, banking frozen-slice positives that failed the evolving test, or climbing scores through outright numerical pathologies. The gates that close those routes are detailed in Sec. II H.

The search engine is MAP-Elites [12], a quality-diversity algorithm that keeps one elite per cell of a behavior archive rather than a single global optimum—preferred over a pure CMA-ES [13] hill-climb, which collapses onto one basin, and over reinforcement learning, which would need dense credit assignment across expensive GPU evolutions with sparse rewards. CMA-ES then warm-starts from the archive champion and refines it locally (Sec. III). Both discovery stages run on a deliberately coarse grid chosen for throughput; production claims are deferred to the high-quality (HQ) replay and resolution matrix (Sec. III).

The headline champion (candidate 146) relies on a phantom scalar sector, so it does *not* bypass the energy-

* shirokov.nm@phystech.edu

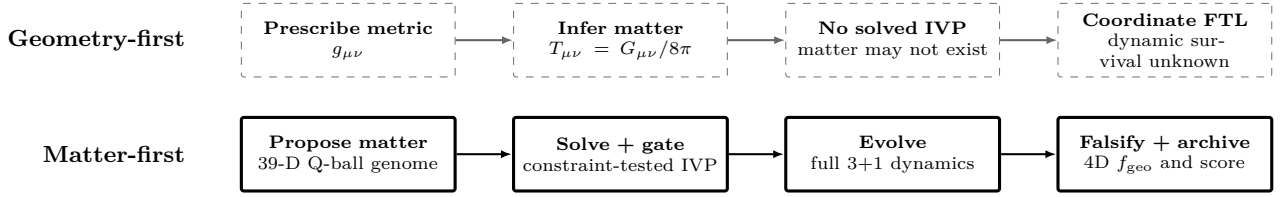


FIG. 1. Geometry-first versus matter-first discovery. Dashed: prescribe a metric and infer its stress-energy. Solid: propose matter, solve and test an initial-value problem (IVP), evolve it, and archive only candidates that pass the 4D null-transport test.

condition violations required of FTL spacetimes [8, 9, 11]. Its value is a demonstration that an automated, matter-first Cauchy pipeline can discover, construct, and dynamically track constraint-satisfying FTL geometries—a necessary first step toward searching the positive-energy sector [14, 15] with the same falsification stack. As a first probe, a matched canonical-only (positive-energy) search within the present ansatz returns $f_{\text{geo}}^{\text{evol}} = 0$ for every accepted evaluation (Sec. VIB).

A. Contributions

Three commitments separate this pipeline from geometry-first studies, and each is a contribution of the paper.

1. **A solved initial-value problem.** Every candidate begins as constraint-clean data from physical scalar sources, not a coordinate script, produced by an end-to-end loop—MAP-Elites proposal, CMA-ES local refinement, constraint solve, GPU evolution, matched scoring—with one explicit reward rule applied to every evaluation (Secs. II–III).
2. **Dynamic feedback.** Full CCZ4 evolution, so dispersing matter collapses the channel and cannot be scored around; claims are promoted through a hierarchical protocol ending in a Richardson resolution ladder on the frozen champion (Sec. III).
3. **4D geodesic falsification.** The headline diagnostic is `ftl_geo_evolving`—null rays integrated through the *time-evolving* metric stack. Frozen-slice geodesics are retained as diagnostics but never as the transport certificate. The surviving candidate’s limits are reported as explicitly as its successes: dispersing exotic matter, a transient ignition controller, and a late-time collapse to an apparent horizon once the exotic support is gone—not passenger transport (Sec. VI).

II. MATTER-FIRST PIPELINE

A. Two-code toolchain

Two open numerical-relativity codes carry the loop. **GRTresna** [16] is a constraint solver: given a matter configuration it returns constraint-satisfying initial data $(\gamma_{ij}, K_{ij}, \phi, \Pi)$ on the initial slice, so that the Hamiltonian and momentum constraints hold before any evolution. **GRTeclyn** [6, 7] is a GPU-accelerated, AMReX-based [17] code that evolves those data forward with the CCZ4 formulation [18] of the Einstein equations under moving-puncture gauge [19, 20]. Neither code is told what metric to produce; the geometry is whatever the proposed matter sources. One evaluation therefore maps a parameter vector θ to a scalar fitness— $\theta \rightarrow$ initial data $\rightarrow$ **GRTeclyn** $\rightarrow$ diagnostics $\rightarrow S(\theta)$ —with post-processing probes reducing the evolution to the fitness that steers the next proposal (Fig. 4).

B. Matter model: canonical and phantom scalar sectors

The matter is built from complex scalar fields with canonical Klein–Gordon dynamics; the production model carries two independent fields, each with a conjugate momentum Π and the sextic potential V of Eq. (6). In the 3+1 split, a complex scalar contributes an energy density ρ and a momentum density S_i that act as the sources in the Hamiltonian and momentum constraints,

$$\rho = \frac{1}{2}\psi^{-4}|\tilde{\nabla}\phi|^2 + \frac{1}{2}|\Pi|^2 + V(\phi), \quad S_i = -\text{Re}(\Pi^*\partial_i\phi), \quad (1)$$

where ψ is the conformal factor and $\tilde{\nabla}$ the conformal spatial gradient. The three terms in ρ are the gradient (spatial shape), kinetic (time variation), and potential (rest) energy of the field, while S_i is nonzero only when the field oscillates in time ($\Pi \neq 0$) with a spatial gradient. Equation (1) is the precise statement of “the matter tells the geometry what to be”: θ sets ϕ and Π , and **GRTresna** returns the (γ_{ij}, K_{ij}) consistent with them.

Because the exotic sign is central to the claim, we state exactly what the `grtresna_bicomplex_scalar` model in-

tegrates. Each complex scalar is canonical, with action

$$S[\Phi] = \int d^4x \sqrt{-g} (-g^{\mu\nu} \partial_\mu \Phi^* \partial_\nu \Phi - V(|\Phi|)), \quad (2)$$

Writing $\Pi = n^\mu \partial_\mu \Phi$ for the conjugate momentum along the normal n^μ , both the canonical field Φ_+ and the exotic field Φ_- obey the *same* 3+1 Klein–Gordon system,

$$\begin{aligned} \partial_t \Phi_\pm &= \alpha \Pi_\pm + \beta^i \partial_i \Phi_\pm, \\ \partial_t \Pi_\pm &= \beta^i \partial_i \Pi_\pm + \alpha \left(K \Pi_\pm + \gamma^{ij} D_i D_j \Phi_\pm - \frac{\partial V}{\partial \Phi_\pm^*} \right) \\ &\quad + \gamma^{ij} \partial_i \alpha \partial_j \Phi_\pm, \end{aligned} \quad (3)$$

with D_i the spatial covariant derivative and $\alpha, \beta^i, \gamma_{ij}$ the usual lapse, shift, and spatial metric. Both sectors are therefore governed by identical right-sign (non-ghost) kinetic operators, free of the gradient and ghost instabilities that afflict a genuine phantom Lagrangian $\mathcal{L} = +\partial_\mu \phi \partial^\mu \phi^* - V$.

The exotic character enters at exactly one place: the *gravitational coupling*. The two sectors add to the Einstein source with opposite sign,

$$G_{\mu\nu} = 8\pi (T_{\mu\nu}[\Phi_+] - T_{\mu\nu}[\Phi_-]), \quad (4)$$

so Φ_- is a negative-energy (ghost-*gravitational*) source while retaining canonical, well-posed field dynamics. Equivalently, this reverses the effective gravitational constant for the exotic sector ($G \rightarrow -G$ for Φ_-); it is a controlled model of the negative-energy source required to violate the null energy condition, not a claim that a fundamental phantom field theory has been made well posed.

The Cauchy problem is unaffected by the flip. Matter enters the CCZ4 evolution equations only through the undifferentiated combinations ρ , S_i , and S_{ij} , so reversing their sign changes lower-order source terms and leaves the principal part—and hence the characteristic structure and strong hyperbolicity of CCZ4 in the moving-puncture gauge [18]—exactly as in the minimally coupled canonical case. The scalar sector, Eq. (3), remains a pair of right-sign wave operators on that background. The coupled system is therefore strongly hyperbolic and locally well posed with the same proof as for ordinary matter. What the sign flip does remove is the positive-energy theorem: no a priori bound controls the long-time behavior of the geometry, which is why every claim below rests on explicit constraint monitoring over a finite window rather than on a stability argument—and why the run ends in gravitational collapse (Sec. VIA).

The word “canonical” therefore describes the equation of motion, not the sign with which each initial lump gravitates. For each lump the optimizer stores $x_{\text{ex},k} \in [0, 1]$, rounded at configuration time to $e_k = \text{round}(x_{\text{ex},k})$ and mapped to $s_k = (-1)^{e_k}$, so there is no fractional interpolation. In the **GRTresna** constraint solve the signed

independent-lump approximation is

$$\rho_{\text{ID}} = \sum_k s_k \rho_k, \quad (S_i)_{\text{ID}} = - \sum_k s_k \sum_{A=1}^2 \Pi_{A,k} \partial_i \phi_{A,k}, \quad (5)$$

with inter-lump stress-tensor cross terms omitted; the profiles are then superposed into the two painted complex fields, one canonical and one phantom. Applying the sign flip consistently to ρ , S_i , and S_{ij} in both the constraint solve and the **GRTeclyn** stress tensor removes the initial-data / evolution mismatch of earlier single-field campaigns. Candidate 146 rounds to $s_k = (+1, -1, -1, +1, -1)$: two canonical and three phantom sources. Because the fields never leave the stable Klein–Gordon sector, the late-time dispersion reported below (Sec. VIA) is *structural*—the finite lifetime of thick-wall Q-balls in an unbound orbital arrangement—rather than a phantom kinetic instability.

C. Q-ball building blocks

The matter is built from Q-balls [21, 22]—localized, non-topological solitons of a complex scalar field whose conserved Noether charge provides self-binding against dispersal. Simpler ansatz families lacked this binding and disperse too rapidly (Sec. VIA). We choose a *complex* rather than real scalar because only a complex field has the phase symmetry that supports a stable charged lump, and its harmonic phase rotation $\phi \propto e^{-i\omega t}$ makes $\Pi = \dot{\phi} \neq 0$, so by Eq. (1) the lump carries a momentum density S_i and can source a nontrivial shift (Sec. VC). Each lump starts from a spherically symmetric equilibrium profile obtained by solving the radial ODE for a sextic potential,

$$V(|\Phi|) = \frac{1}{2} m^2 |\Phi|^2 - \frac{1}{4} \lambda |\Phi|^4 + \frac{1}{6} \mu |\Phi|^6, \quad (6)$$

with fixed parameters $m = 1$, $\lambda = 640$, $\mu = 85333$, $\omega = 0.8$. The attractive quartic ($\lambda > 0$) provides self-binding; the repulsive sextic ($\mu > 0$) is *required* for a stable three-dimensional Q-ball—a pure quartic is the critical case and either collapses or disperses. The values are fixed by requirements rather than tuned: $m = 1$ sets the mass (hence length) unit; λ and μ place the stable thick-wall window $\omega_{\min} < \omega < m$ comfortably around the working frequency (with λ^2/μ setting $\omega_{\min} \approx 0.316$); and $\omega = 0.8$ keeps the equilibrium profile inside a radius of ~ 7 code units so several lumps fit in the box. These microphysical constants are not part of the search.

D. Shell Parameterization

The search vector has 39 dimensions because the arrangement of the lumps, not their microphysics, is what is optimized. The shell contains five lumps—the maximum

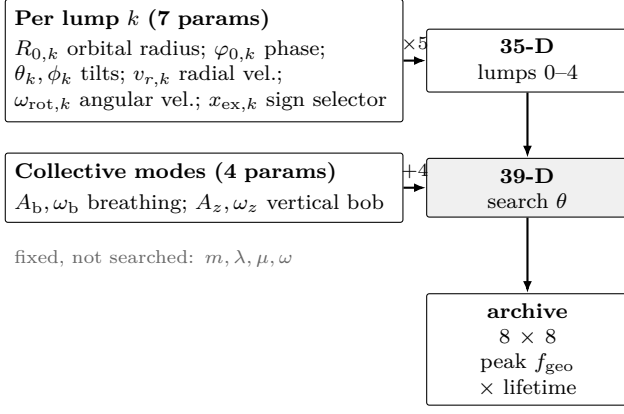


FIG. 2. Search parameterization of the trajectory Q-ball shell: $5 \times 7 = 35$ per-lump placement parameters plus 4 collective modes give the 39-D genome θ . MAP-Elites stores one elite per cell of an 8×8 archive indexed by peak shortcut and FTL lifetime. Q-ball microphysics is held fixed.

supported by the solver’s state layout—giving enough independent tilted orbits to build a structured source while keeping the search space tractable. Each lump carries seven placement parameters—orbital radius R_0 , phase, two tilt angles, radial velocity, angular velocity, and a source-sign selector—for $5 \times 7 = 35$ dimensions, and four collective variables set the shell’s breathing and vertical-oscillation amplitudes and frequencies, giving 39 in total (Fig. 2). The sign variable is sampled continuously but applied as the binary selector above, letting the optimizer choose which lumps source positive and which negative stress-energy.

E. Lump orbit parameterization

Each lump moves in its own tilted orbital plane. The support center of lump k is prescribed by

$$r_k(t) = \max[r_{\min}, R_{0,k} + v_{r,k}t + A_b \sin(\omega_b t)],$$

$$\varphi_k(t) = \varphi_{0,k} + \omega_{\text{rot},k}t, \quad z_b(t) = A_z \sin(\omega_z t), \quad (7)$$

followed by rotation into the laboratory frame through the two tilt angles (θ_k, ϕ_k) . Thus each lump spirals radially (drift $v_{r,k}$), rotates in its own plane (angular velocity $\omega_{\text{rot},k}$; negative values are retrograde), and shares a collective radial breathing (A_b, ω_b) and vertical oscillation (A_z, ω_z). The range of possible orbits includes retrograde and prograde configurations, fast and slow rotation, radially drifting spirals, and nearly static scaffolds.

F. Trajectory controller (PD pump)

During evolution, a proportional–derivative (PD) controller drives the two real components of each complex field toward a moving target profile at the prescribed

support center:

$$\partial_t \Pi_A|_{\text{PD}} = -wk_p(\phi_A - \phi_A^*) - wk_d(\Pi_A - \Pi_A^*), \quad A = 1, 2, \quad (8)$$

with gains $k_p = 12$, $k_d = 7$, and w the product of a localized envelope and a constraint-sensitive governor. This is an external, non-Lagrangian source in the scalar momentum equations; no corresponding pump stress-energy is added to $T_{\mu\nu}$, so $\nabla_\mu T^{\mu\nu} = 0$ is not exact while the pump is active. The volume-integrated pump power $P = \int \alpha (\Pi \cdot S_\Pi) \sqrt{\gamma} dV$ recorded in `collapse_diagnostics.dat` integrates to $E_{\text{pump}} \simeq 1.6 \times 10^{-17}$ over the ignition window of the pump-on HQ run—consistent with floating-point noise relative to the matter amplitude ($|\phi|_{\text{max}} \sim 7 \times 10^{-2}$). The pump therefore acts as a soft trajectory guide rather than a large energy injector. The campaign default switches it off at $t_{\text{pump}} = 4$, and scoring credits 4D geodesic transport only for $t_{\text{emit}} \geq t_{\text{pump}}$, so the headline geodesic—which launches at exactly $t_{\text{emit}} = 4$ —samples a freely evolved metric ($\nabla_\mu T^{\mu\nu} = 0$). A dedicated pump-free HQ twin with $t_{\text{pump}} = 0$ (Sec. V H) tests whether the shortcut survives with the PD term identically off from $t = 0$.

G. Constraint gate and source-model consistency

A post-load constraint gate rejects inconsistent initial data before any GPU time is spent: the solved data are loaded into `GRTEclyn` for a short verification integration, and the volume-weighted L^2 constraint norms must stay below a campaign threshold or the candidate is discarded and logged. The bicomplex Q-ball campaigns use 3×10^{-2} for the Hamiltonian and 10^{-2} for the momentum residual; candidate 146 clears both comfortably at $\|\mathcal{H}\|_2 = 5.9 \times 10^{-3}$ and $\|\mathcal{M}\|_2 = 1.1 \times 10^{-4}$. This gate, together with bounded CCZ4 constraint growth, establishes numerical control and source-model consistency for the `grtresna_bicomplex_scalar` model of Sec. II B.

H. Scoring: how every evaluation is rewarded

One fixed rule reduces every evaluation to a single fitness $S(\theta)$. Each named quantity below is a normalized diagnostic in $[0, 1]$ (penalties are negative) and the score is their weighted sum,

$$S = 1000 f_{\text{geo}}^{\text{evol}} + 1000 f_{\text{geo}}^{\text{op}} + 600 P_{\text{ftl}} + 200 F_{\text{op}} + 200 C_{\text{curv}} + 5 G_{\text{nt}} + g_h \Sigma_{\text{health}} + 40 w_x X_{\text{exotic}} + 200 H_{\text{hor}}, \quad (9)$$

with the health block

$$\Sigma_{\text{health}} = 150 \text{ surv} + 60 E + 40 \text{ stab} + 30 \text{ cmv} + 10 \mathcal{H}_{\text{ok}} + 15 I, \quad (10)$$

gated by the nontriviality factor $g_h \in [0, 1]$ so that a flat configuration earns no health reward. Inside the

TABLE I. Reward terms in the campaign objective [Eq. (9)], applied identically to every MAP-Elites evaluation.

symbol	rewards / penalizes	weight
$f_{\text{geo}}^{\text{evol}}$	4D evolving null shortcut	1000
$f_{\text{geo}}^{\text{op}}$	geodesic (frozen) shortcut	1000
P_{ftl}	FTL persistence (gated)	600
F_{op}	operational coord. FTL (gated)	200
C_{curv}	curvature activity	200
G_{nt}	nontrivial geometry	5
surv	structural survival	$150 g_h$
E	energy-condition health	$60 g_h$
stab	metric stability	$40 g_h$
cmv	co-moving stability	$30 g_h$
I	instability penalty	$15 g_h$
$\mathcal{H}_{\text{ok}}$	constraint health	$10 g_h$
X_{exotic}	exotic-matter penalty	$40 w_x$
H_{hor}	horizon-formation penalty	200

score the raw advantage f_{geo} is mapped to a saturating magnitude $\hat{f} = \min[(f_{\text{geo}} - 10^{-3})/(0.2 - 10^{-3}), 1]$, so marginal shortcuts earn proportionally less and full marks require $f_{\text{geo}} \geq 20\%$; the two geodesic slots then pay $1000 \hat{f} p_s$ with p_s the structural persistence (density retention $\times$ coherence $\times$ confinement). Whenever a 4D evolving trace exists it takes precedence: the frozen slot is zeroed, $f_{\text{geo}}^{\text{op}} \equiv 0$, so the two 1000-weight terms are mutually exclusive rather than additive. Table I lists what each term measures. Two design choices matter. First, transport dominates and every transport-flavored term— $f_{\text{geo}}^{\text{evol}}$, P_{ftl} , and F_{op} —is multiplied by the fraction of matter still confined, so no candidate can bank a coordinate superluminal channel after its lumps have flown apart. Second, the exotic penalty $X_{\text{exotic}} < 0$ charges for energy-condition violation [23] but is deliberately weak: at the campaign value $w_x = 0.2$ and the graded floor $X_{\text{exotic}} \geq -1.6$ it reaches at most ≈ 13 points against the ≈ 417 the champion earns from the persistence-gated geodesic term. The optimizer may therefore spend exotic matter freely in exchange for genuine transport, which is why the archive populates the phantom sector rather than being pinned to $X_{\text{exotic}} = 0$. Search-tier scores are values of S on a $t = 16$ evolution (Sec. III); the only production S values appear in the pump-free comparison (Table VIII). For the CMA-ES champion (candidate 146) the search-tier score is $S = 852.15$ (Sec. III).

I. Search and validation stages

MAP-Elites and CMA-ES evaluate $\mathcal{O}(10^2)$ genomes, so the discovery loop uses a reduced GRTEclyn configuration for wall-clock speed (Table II). Archive and CMA scores are therefore *search-tier* diagnostics, not continuum claims; only the frozen champion is re-run at pro-

TABLE II. GRTEclyn evolution settings: search tier (MAP-Elites / CMA-ES) versus production validation (HQ / Richardson medium). Search uses a reduced grid for throughput; validation upgrades L , N , **max_level**, and t_{stop} . Campaign env: GRTRESNA_EVOLUTION_{L,N}_FULL, _MAX_LEVEL, _REGRID_THRESHOLD.

parameter	search	HQ / RM
L	64	128
N	128	256
$h = L/N$	0.5	0.5
max_level	1	3
regrid_threshold	0.1	0.02
t_{stop}	16	30

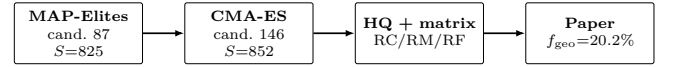


FIG. 3. Outer campaign ladder. MAP-Elites finds the diverse archive and champion (87); CMA-ES refines it to candidate 146; only the refined champion enters high-resolution replay and the resolution matrix.

duction resolution. The campaign ladder (Fig. 3) has four stages.

- **Stage 1 — MAP-Elites discovery.** Quality-diversity search on the search grid, with per-lump gravitational signs retained in evolution. The coarse AMR is intentional: it buys enough evaluations to fill the behavior archive.
- **Stage 2 — CMA-ES refinement.** Warm-start from the MAP-Elites champion and its nearest elites; local covariance-matrix adaptation at the *same* grid, t_{stop} , and scoring rule. Only the refined top genome is promoted.
- **Stage 3 — production replay (HQ).** The frozen champion is re-solved and evolved at higher fidelity and longer t_{stop} , with multiple emission times and multi-ray geodesic probes. Every headline number in this paper comes from here, not from MAP-Elites.
- **Stage 4 — resolution and domain matrix.** Matched coarse/medium/fine runs (RC/RM/RF at fixed $L = 128$, $N = 192/256/384$) test continuum stability; matched-spacing domain runs (DS/DL at $h = 0.500$, $L = 96/160$) test boundary sensitivity on the same t_{stop} window.

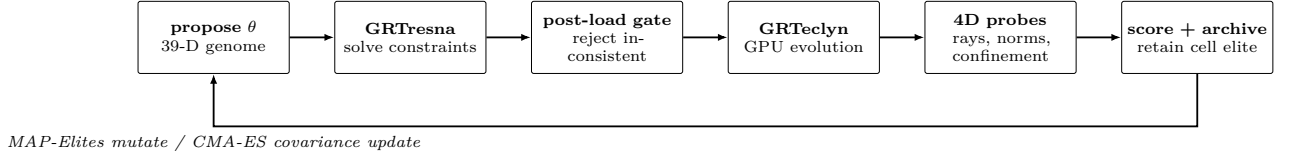


FIG. 4. Inner evaluation loop shared by MAP-Elites and CMA-ES. Failed constraint solves and post-load checks never enter GPU evolution; successful evaluations compete for archive cells (MAP-Elites) or update the CMA distribution.

III. SEARCH METHODOLOGY

A. MAP-Elites quality-diversity

MAP-Elites [12] is a quality-diversity search: instead of returning one champion, it maintains an 8×8 archive with at most one elite per behavior cell. A candidate displaces the occupant of its cell only if its score $S(\theta)$ [Eq. (9)] is higher. The two archive axes are peak short-cut strength and FTL lifetime, so the campaign can keep short-lived strong channels and longer weaker ones side by side. The bicomplex campaign scheduled 200 evaluations: 148 completed GPU evolution, 50 failed the post-load gate, one failed the GRTresna admission gate, and one failed the constraint solve. The best score rose from 690.1 to 825.19 and last improved at batch 10, when candidate 87 entered the strongest cell (Fig. 5a). The fixed 200-evaluation budget is set to outlast saturation—termination requires a plateaued archive-best score and static cell coverage—and here both stall by batch 10, so the budget and the saturation criterion coincide.

Nine of the 64 cells ended occupied (14.06% coverage): all eight bins of the strength axis in the highest-lifetime row, plus one zero-lifetime cell. The archive is therefore a clean strength ladder—peak f_{geo} climbs from 1.8% to 25.4% while every elite holds its channel open for the full scored window—and score broadly tracks it, because transport carries the 1000-weight terms of Eq. (9) and nothing else in the objective comes close. What score does *not* track is persistence, which wanders between 27.8% and 39.6% with no trend along the ladder—the first sign that strength and confinement are set by different features of the genome.

B. The strongest shortcut is not the promoted one

Neither champion is the largest shortcut its stage produced (Table III). The strongest evolving advantage anywhere in the MAP-Elites run is 34.56% (eval 99), nine percentage points above the archive champion; the strongest in the whole study is 39.46% (CMA-ES eval 136), nearly double candidate 146. Both were rejected by the same mechanism. Eval 99 retained 9.7% of its matter and eval 136 22.5%, against 39.6% and 41.7% for the promoted genomes, and because every transport-flavored term in Eq. (9) is multiplied by that retained fraction, their scores land at 314 and 534 instead of 825

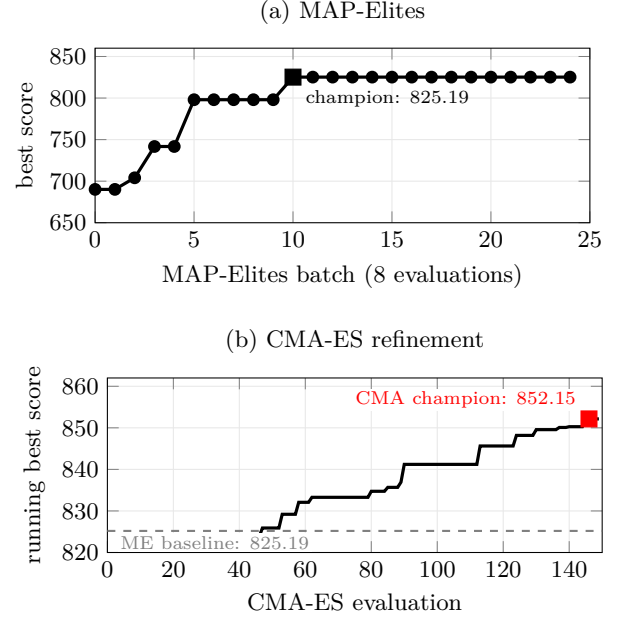


FIG. 5. Search progress. (a) MAP-Elites running best score over 200 evaluations; the champion (candidate 87) enters cell (7, 7) at batch 10 with $S = 825.19$. (b) CMA-ES refinement over 150 evaluations, warm-started from MAP-Elites champion 87; dashed line marks the MAP-Elites best (825.19); the running best reaches 852.15 at candidate 146.

TABLE III. Strongest raw shortcut of each search stage against the genome actually promoted. The first four rows are search tier, the last is the production replay; persistence is the confinement-gated structural persistence. The promoted champions are weaker and better confined.

stage	role	f_{geo}	persist.	S
MAP-Elites	strongest (99)	34.56%	9.7%	314.0
MAP-Elites	champion (87)	25.35%	39.6%	825.2
CMA-ES	strongest (136)	39.46%	22.5%	534.3
CMA-ES	champion (146)	20.12%	41.7%	852.1
HQ RM	replay of 146	20.21%	—	—

and 852. The gate is doing exactly what Sec. IIH designed it to do: a large f_{geo} measured while the source is flying apart buys almost nothing.

The trade is systematic, not anecdotal. Over the 119 MAP-Elites evaluations with $f_{\text{geo}} > 0$, strength and

persistence are anti-correlated ($r = -0.15$), and their strongest quartile averages 27.3% persistence against 31.0% for the weakest; the 141 such CMA-ES evaluations are more strongly anti-correlated still ($r = -0.32$). Reading down Table III, the campaign is already sketching the same Pareto frontier the geometry-first atlas maps kinematically in Sec. VID: the configurations that open the widest channel are the ones least able to hold it open.

C. CMA-ES refinement and the promotion of candidate 146

CMA-ES warm-started from candidate 87 and ran 150 evaluations (144 GPU-ok) at the same search grid, reaching $S = 852.15$ at candidate 146 ($\Delta S \approx +27$; Fig. 5b). The gain is *entirely* on the persistence side: the search-tier f_{geo} fell from 25.35% to 20.12% while structural persistence rose from 39.6% to 41.7%. CMA-ES bought that by collapsing onto a quasi-static scaffold—all five angular velocities below 5×10^{-3} and breathing amplitude 0.083—which keeps the lumps in place long enough for the gated terms to pay out. The run is converged rather than merely lucky: the eight best genomes all sit at $f_{\text{geo}} = 19.9\text{--}20.4\%$ and persistence 41.5–41.7%, within 4 points of score of each other, so candidate 146 is the representative of a stable basin and not an isolated spike.

Candidate 146 then goes to production replay, where the $L = 128$, $N = 256$ evolution reproduces the search-tier shortcut at 20.21% with all five rays reaching—agreement to 0.5% relative across a doubling of resolution and a tripling of refinement depth. Its 41.7% retention, however, is measured at $t = 16$, and the production window runs to $t = 30$: the longer integration stresses persistence much harder (Sec. VIA).

IV. DIAGNOSTICS AND PROBE VALIDATION

A. The 4D evolving null-geodesic trace

The transport claim rests on tracing light rays through the simulated spacetime, not on reading coordinate speeds off a slice. A coordinate operational shortcut F_{op} [Eq. (12)] can flag tilted light cones, but it is partly gauge dependent and can be inflated by a coordinate choice. The authoritative probe therefore integrates the null-geodesic equations

$$\dot{x}^\mu = g^{\mu\nu} k_\nu, \quad \dot{k}_\mu = -\frac{1}{2}(\partial_\mu g^{\alpha\beta}) k_\alpha k_\beta, \quad (11)$$

for a photon with position x^μ and momentum k_μ . “4D” means the ray is not confined to one snapshot: it is pushed through a stack of stored time slices, so at each step the metric $g_{\mu\nu}(t, \mathbf{x})$ used is the one that actually exists when the photon is at that point. This distinguishes it from a *frozen* trace, which wrongly assumes a single slice persists for the whole flight.

The construction is as follows. A ray is emitted at time t_{emit} from a fixed source point toward a fixed receiver. The quantities that enter the shortcut formulas are *transit durations*, not absolute grid arrival times: t_B^{curved} (respectively t_{min}) is the coordinate time elapsed along the curved path after emission, and t_B^{flat} (respectively t_{flat}) is the matched flat-space flight duration between the same endpoints. Absolute arrival on the grid is $t_{\text{arrival}} = t_{\text{emit}} + t_B^{\text{curved}}$. The fractional advantage is

$$F_{\text{op}} = \frac{t_B^{\text{flat}} - t_B^{\text{curved}}}{t_B^{\text{flat}}}, \quad (12)$$

$$f_{\text{geo}} = \max\left(0, \frac{t_{\text{flat}} - t_{\text{min}}}{t_{\text{flat}}}\right), \quad (13)$$

so $f_{\text{geo}} = 0$ means no shortcut. A launch is accepted only if its rays stay physical: the relative drift in the null Hamiltonian $g^{\mu\nu} k_\mu k_\nu = 0$ must remain below 10^{-2} , which rejects rays that have accumulated interpolation error. Because a ray needs a long remaining segment of evolved geometry to reach the far endpoint (flat transit is $t_{\text{flat}} = 14.4$) while the metric stack ends at $t = 30$, late launches cannot finish a fair comparison. We therefore sweep seven early launch times, $t_{\text{emit}} = 0, 2, \dots, 12$ (five rays each), and report the largest accepted f_{geo} . Across the pump-on validation matrix the peak sits at $t_{\text{emit}} = 4$, just after the PD pump switches off, and has fallen to zero by $t_{\text{emit}} = 12$, so the window brackets the full life of the channel. (In the pump-free twin the optimum sits at the edge of the same window; Sec. VH.)

B. Freely falling observer timing metric

As a stronger complement to the fixed-plane coordinate diagnostic f_{geo} , we define an observer-to-observer timing metric. At the initial slice, an emitter A and receiver B are placed at the central source and detector locations used by the null-ray probe and are released with four-velocity equal to the future-directed Eulerian normal n^μ . Their worldlines γ_A and γ_B are then evolved as timelike geodesics,

$$\frac{dx^\mu}{d\tau} = g^{\mu\nu} p_\nu, \quad \frac{dp_\mu}{d\tau} = -\frac{1}{2}\partial_\mu g^{\alpha\beta} p_\alpha p_\beta, \quad g^{\mu\nu} p_\mu p_\nu = -1, \quad (14)$$

where τ is the observer’s proper time. Emission is scheduled by the emitter’s clock, $\tau_A = \tau_{\text{emit}}$, rather than by simulation coordinate time. At the emission event, an orthonormal spatial triad $e_{(a)}^\mu$ is constructed in the emitter’s rest frame and the photon wave vector is launched as

$$k^\mu = u_A^\mu + \hat{n}^a e_{(a)}^\mu, \quad \delta_{ab} \hat{n}^a \hat{n}^b = 1, \quad (15)$$

up to an irrelevant positive affine rescaling. The two launch angles and the arrival affine parameter are solved

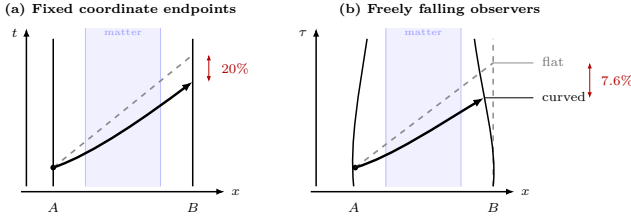


FIG. 6. Two operational measurements of the same candidate 146 (RM). (a) Fixed endpoints: dashed = flat ray, solid = curved ray arriving 20% earlier at fixed B . (b) Free-fall observers: both receivers start at B ; the solid worldline drifts in curved spacetime while the dashed one stays inertial. The solid photon hits the curved receiver earlier (“curved”) than the dashed photon hits the inertial one (“flat”). The 7.6% bracket is the proper-time advance. The two protocols differ in what they measure, not in candidate quality.

simultaneously so that the ray intersects the moving receiver worldline γ_B : reception is a worldline event, not the crossing of a coordinate detector plane. The physical miss distance $\delta_B = (\gamma_{ij}\Delta x^i\Delta x^j)^{1/2}$ at the candidate event is retained as a quality diagnostic alongside the null and timelike mass-shell residuals.

For the matched Minkowski experiment, the two inertial observers are initially comoving, their clocks are synchronized on the reference initial slice, and their separation is the initial proper length

$$D_0 = \int_A^B \sqrt{\gamma_{ij} dx^i dx^j}. \quad (16)$$

The flat receiver therefore records $\tau_{B,\text{flat}} = \tau_{\text{emit}} + D_0$, and if the curved reception event occurs at receiver proper time $\tau_{B,\text{curved}}$ the metric is

$$f_{\text{ff}} = \frac{\tau_{B,\text{flat}} - \tau_{B,\text{curved}}}{D_0}. \quad (17)$$

A positive f_{ff} means the freely falling receiver records the photon earlier than in the matched flat experiment. Absorbing endpoint lapse, shift, clock rate, and receiver motion into the measured worldlines makes this less coordinate-conditioned than Eq. (13), though not fully gauge invariant: the initial slice and the endpoint-selection prescription remain part of the operational definition.

For candidate 146 in the corrected RM replay, the fixed-endpoint probe gives $f_{\text{geo}} \simeq 20\%$, whereas the freely falling protocol gives $f_{\text{ff}} = 7.62\%$: $D_0 = 13.851$, $\tau_{B,\text{flat}} = 17.851$, and $\tau_{B,\text{curved}} = 16.796$, so the receiver-clock advance is 1.055 and $f_{\text{ff}} = 1.055/13.851 = 7.62\%$. The smaller magnitude is not a degradation of the candidate. The two protocols measure different things (Fig. 6): releasing the endpoints changes the reception event, and proper-time readout absorbs their motion and accumulated clock rates. Which piece of geometry accounts for the difference is settled quantitatively in Sec. V C. Normalizing by D_0 rather than by the absolute clock read-

ing $\tau_{B,\text{flat}}$ makes the fraction independent of an arbitrary common shift of the clocks’ zero.

Moving-puncture coordinates are prone to slice stretching, so it matters that this protocol never leaves the weak field. Both A and B start in the near-flat region (Table VI) and are released with Eulerian normal four-velocity; the reported quantity is the receiver’s own proper time at a reception event on its own worldline, normalized by an initial proper length. The 7.6% advance is therefore physically unambiguous in a way the coordinate figure is not.

C. Null-constraint accuracy along the flight path

The 10^{-2} acceptance gate is a conservative safety bound, not the achieved accuracy. Two mechanisms keep the ray far tighter. First, whenever the absolute null Hamiltonian exceeds 10^{-6} the momentum is re-projected onto the local null cone before the next Runge–Kutta step, so interpolation error cannot accumulate secularly. Second, the normalized null constraint stays small between projections. Along the headline candidate-146 ray ($t_{\text{emit}} = 4$, RM), the gauge-independent ratio $k_\mu k^\mu / (k^t)^2$ has median 3.0×10^{-4} (a factor ~ 30 below the gate) and a brief peak of 4.1×10^{-3} in the strong-field core (still a factor ~ 2.4 under the gate). The ray therefore stays null to the stated tolerance throughout and does not drift toward a timelike or spacelike trajectory. The complementary relative-drift diagnostic recorded by the probe, $h_{\text{rel}}^{\text{max}} = 2.19 \times 10^{-3}$ in RM, tells the same story.

D. Probe controls and calibration

Three controls bracket the probe’s dynamic range before any candidate is evaluated.

- **Minkowski (flat):** $f_{\text{geo}} = 0$ by construction; verifies that the tracer introduces no spurious shortcut on trivial data.
- **Prescribed Alcubierre [3]:** a moving-bubble metric ($v_s = 0.5$, bubble radius 2, wall width 2) is built analytically and fed to the evolving tracer, yielding $f_{\text{geo}} \approx 32\%$ with 5/5 rays reaching. This confirms that the 4D probe recovers a known coordinate shortcut.
- **Negative control (eval086):** an earlier campaign candidate whose frozen-slice diagnostic shows $f_{\text{geo}} \approx 5.8\%$, but whose 4D evolving HQ trace collapses to 0% (0/5 rays). This false positive motivated the design rule that the evolving trace is authoritative: frozen peaks alone do not certify transport.

The positive campaign control is the CMA-ES champion (candidate 146): every accepted launch in all three

resolution-matrix runs has 5/5 rays reaching and passes the null-Hamiltonian gate.

V. RESULTS: CANDIDATE-146 WARP PRECURSOR

A. Provenance and genome

Candidate 146 is the CMA-ES-refined champion from the bicomplex campaign (Sec. III). Its provenance freezes the exact 39-dimensional genome, source checksums, solver parameters, and source commit. The finest run (RF) additionally stored field maps, from which the snapshots of Fig. 8 are drawn.

B. Matter dynamics

Candidate 146 uses five lumps with source signs $s_k = (+1, -1, -1, +1, -1)$ (two canonical, three phantom; Sec. IIB). The collective breathing is small ($A_b = 0.083$, $\omega_b = 0.429$), the vertical oscillation is moderate ($A_z = 2.32$, $\omega_z = 0.016$), and the initial radii are $R_0 = (3.69, 5.50, 2.83, 5.59, 2.61)$. All angular velocities are negative, $\omega_{\text{rot}} = (-0.0017, -0.0007, -0.0031, -0.0009, -0.0049)$, so all five trajectories are retrograde but extremely slow. Radial drifts are small [$v_r = (-0.060, -0.010, 0.001, -0.005, -0.006)$] and roughly an order of magnitude quieter than MAP-Elites champion 87, the quasi-static scaffold CMA-ES converged to (Sec. III).

The PD trajectory controller (Sec. IIF) drives the lumps during the ignition phase ($t < 4$) and then switches off; the evolution remains numerically controlled throughout (Fig. 10). Figure 7 shows an unscaled top-down (x - y) projection confirming that the shell spans only a few code units across the null-probe corridor. Those support centers are the intended dynamical scaffold, not tracked density isosurfaces, and the field follows them imperfectly: confinement falls from 72% to 11% and rms spread grows by a factor 3.27 by $t = 30$. Candidate 146 is a transient, initially driven configuration, not a bound five-body soliton.

C. Transport mechanism: conformal contraction, not frame dragging

The obvious candidate mechanism is frame dragging. Each Q-ball lump oscillates with $\Pi \neq 0$, so by Eq. (1) it carries a momentum density S_i that sources a shift β^i , and a shift tilts the local light cones—which is how the Alcubierre family transports. The shift is certainly there: it vanishes identically on the initial slice, and the moving-puncture gauge then grows it out of S_i and A_{ij} until the peak corridor value $|\beta^x|$ reaches 4.3×10^{-2} at the headline

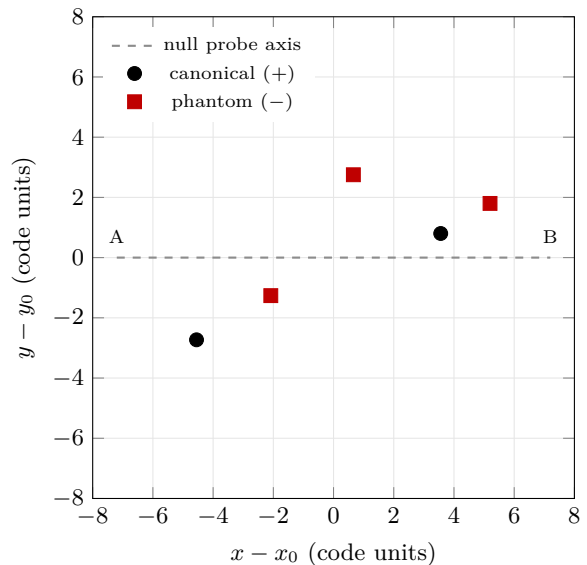


FIG. 7. Top-down (x - y) projection of the candidate-146 initial lump centers at true grid scale (box center $(x_0, y_0) = (64, 64)$; no magnification). Black circles are the two canonical ($+\rho$) lumps, red squares the three phantom ($-\rho$) lumps [signs $(+, -, -, +, -)$]. The dashed line is the x -directed null-probe corridor from emitter A to receiver B ($|x - x_0| \leq 7.2$). The five solitons occupy a region a few code units across, comparable to the corridor half-length.

launch, an order of magnitude above the endpoint values of Table VI. But it is bipolar—the corridor spans $\beta^x \in [-2.0, +4.3] \times 10^{-2}$, the visual signature in the bottom row of Fig. 8—and that matters more than its amplitude.

To attribute the timing gain we re-trace the headline five-ray fan through the stored metric stack with one ADM field at a time replaced by its flat value (Table IV). These are kinematic substitutions on the stored space-time, not new constraint solutions; they apportion the measured f_{geo} rather than describe alternative universes. The result inverts the naive expectation. Deleting the shift entirely *improves* the shortcut, $20.2\% \rightarrow 22.8\%$: because β^x changes sign along the corridor, the segments that speed a $+x$ ray are more than cancelled by those that slow it, so net frame dragging costs about 2.5 percentage points, and a pure-shift geometry transports no faster than Minkowski at all. The other two fields carry the effect: flattening the lapse collapses f_{geo} to 1.7%, and the spatial metric alone already delivers 6.8%.

Both surviving contributions trace to the phantom sector rather than to the matter current. The negative-energy lumps contract the corridor, holding its proper length below the coordinate separation 14.4 throughout the flight (Sec. VE), and they raise the clock rate along it: the corridor-averaged lapse, 0.955 at launch, crosses unity at $t \approx 6.5$ and peaks at 1.34 mid-transit. Candidate 146 is therefore a *conformal-well precursor*: the exotic matter digs a shortened, fast-clock corridor, and the shift it inevitably also sources is a mild parasitic drag.

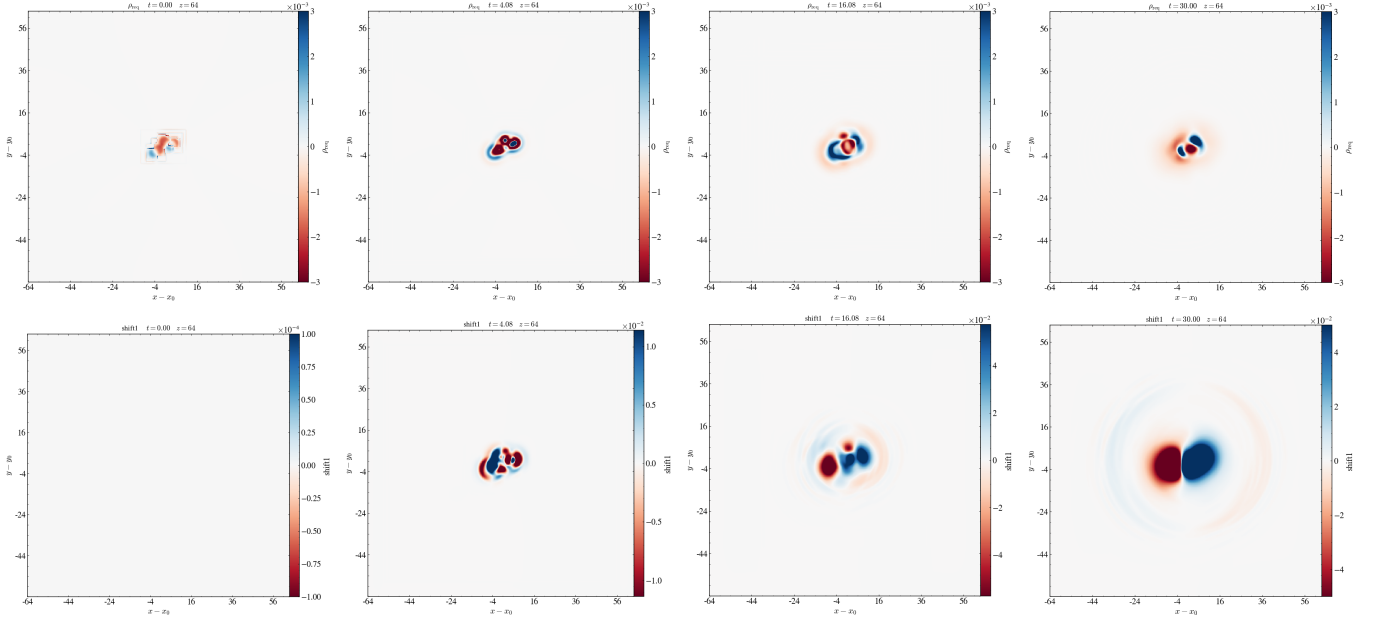


FIG. 8. Evolving dynamics of candidate 146 in the finest run (RF, $N = 384$), mid-plane ($z = z_0$) slices at $t = 0, 4, 16, 30$. *Top row*: required energy density ρ_{req} , with blue the canonical ($+\rho$) lumps and red the phantom ($-\rho$) lumps—a direct view of the negative-energy sector of Sec. II B. *Bottom row*: the shift component β^x along the null-probe axis. Its bipolar structure is why frame dragging does not carry the shortcut (Table IV); the transporting geometry is the conformal well dug by the phantom lumps in the top row. The channel is strongest just after pump-off ($t = 4$, the peak- f_{geo} launch); by $t = 30$ both the matter and the shift structure have spread and weakened, consistent with the transient nature of the shortcut.

TABLE IV. Mechanism ablation for candidate 146: peak evolving f_{geo} when one ADM field of the stored metric stack is replaced by its flat value, with the same five-ray fan and launch $t_{\text{emit}} = 4$. Substitutions are diagnostic, not constraint-satisfying. Removing the shift *raises* the shortcut; removing the lapse elevation removes almost all of it. Baseline re-traces of the archived stacks give 20.15% (RM) and 20.60% (RC), within 0.2 percentage points of Table VII. Source: `mechanism_ablation.txt`.

metric retained	f_{geo} (RM)	f_{geo} (RC)
all (baseline)	20.15%	20.60%
$\beta^i \rightarrow 0$ (no frame dragging)	22.76%	23.10%
$\alpha \rightarrow 1$ (no lapse elevation)	1.70%	1.63%
γ_{ij} only ($\alpha = 1, \beta^i = 0$)	6.83%	6.96%
β^i only ($\alpha = 1, \gamma_{ij} = \delta_{ij}$)	0.00%	0.00%

This also settles what would otherwise be a loose end. The spatial-metric-only entry, 6.8% (RM) and 7.0% (RC), lands at the lower edge of the freely falling observer band $f_{\text{ff}} = 7.0\text{--}7.7\%$ measured independently in Sec. IV B, because the free-fall receiver reports the proper-distance contraction and discards the slicing-dependent lapse elevation—exactly the ~ 13 -point remainder of the coordinate figure. So 20.2% is the coordinate-referenced advantage in the selected gauge, 7.6% is the observer-level one, and the ablation says which piece of geometry supplies each.

TABLE V. Evolving null-geodesic transport certificate for candidate 146 in the headline RM run ($L = 128, N = 256, t = 30$).

quantity	value
peak 4D f_{geo}	20.21%
frozen-peak f_{geo}	29.45%
rays reached	5/5
t_{emit}	4.00
t_{arrival}	15.49
$t_{\text{min}} = t_{\text{arrival}} - t_{\text{emit}}$	11.49
t_{flat}	14.40
t_{transit}	14.40
$h_{\text{quality_ok}}$	true ($h_{\text{rel}}^{\text{max}} = 2.19 \times 10^{-3}$)
search-tier peak (CMA-ES)	20.12% (3/3 rays)

D. 4D falsification certificate

The headline claim is operational: in the RM run, null rays between the fixed endpoints arrive earlier than in flat space. Table V lists the quantities reported by the evolving-geodesic probe.

Explicitly, $(t_{\text{flat}} - t_{\text{min}})/t_{\text{flat}} = (14.40 - 11.49)/14.40 = 20.21\%$, matching Eq. (13). The endpoint timing corresponds to an effective $c/(1 - f_{\text{geo}}) \simeq 1.25c$ relative to the matched flat path; nothing physical moves outside its local light cone (Fig. 6a).

The frozen peak in the same table reaches 29.45%,

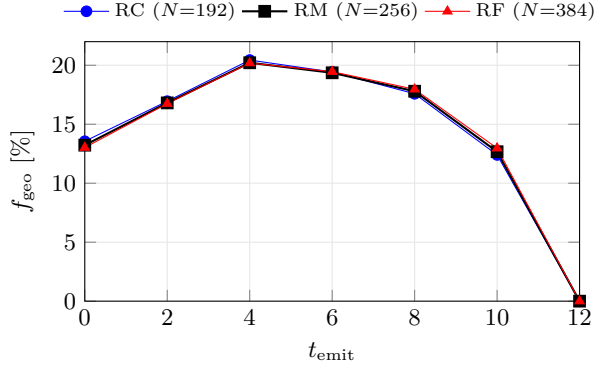


FIG. 9. Evolving null-geodesic emission sweeps ($t_{\text{emit}} = 0, 2, \dots, 12$) for all three resolution-matrix runs; the metric evolves to $t = 30$, but late launches lack enough remaining geometry for a complete transit. All runs peak at $t_{\text{emit}} = 4$, just after PD pump-off, and agree within 0.26 percentage points. Every launch has 5/5 rays reaching and passes the null-Hamiltonian quality gate; at $t_{\text{emit}} = 12$ the rays arrive no earlier than the flat baseline, so the clamped advantage [Eq. (13)] is zero.

some nine percentage points higher, and the gap is physical rather than an error bar: the frozen trace assumes one snapshot persists for the whole flight, whereas the 4D trace watches the channel open and then weaken while the photon is inside it. Only the latter certifies transport. The frozen peak is itself stable across the resolution ladder (29.1–29.5%), so the gap is not a snapshot-quality artifact. Sweeping the launch time (Fig. 9) shows the evolving advantage peaking at $t_{\text{emit}} = 4$ in all three runs and decaying monotonically thereafter, so successful transport coexists with the dispersal of Sec. VIA only inside a finite window.

E. Endpoint gauge and proper-time invariance

The timing advantage of Eq. (13) compares two *coordinate* transit durations between fixed endpoints, which is only a physical statement if those endpoints behave as nearly stationary, nearly flat observers—a deep lapse well ($\alpha \ll 1$) there could manufacture a coordinate advance with no proper-time content. The emitter and receiver sit on the propagation axis at $A = (56.8, 64, 64)$ and $B = (71.2, 64, 64)$ in the $L = 128$ box centered on $(64, 64, 64)$, a corridor of coordinate length $t_{\text{flat}} = 14.4$. At the headline launch $t_{\text{emit}} = 4$ both are close to flat and stationary (Table VI): $\alpha_A = 0.96$, $\alpha_B = 0.99$, and $|\beta^i| \lesssim 3.5 \times 10^{-2}$. The lapse never enters the dangerous $\alpha \ll 1$ regime; at late times it drifts modestly *above* unity ($\alpha \lesssim 1.2$), which if anything makes the coordinate advantage conservative, since a stationary clock with $\alpha > 1$ accumulates proper time faster than coordinate time.

As a gauge-robustness check, we re-express the advantage in the proper time of a static receiver at B , whose clock ticks at $d\tau = \sqrt{-g_{00}(B)} dt$. Integrating the mea-

TABLE VI. Lapse α and shift magnitude $|\beta^i|$ at the fixed probe endpoints A (emitter) and B (receiver) in RM, sampled through the emission window. The endpoints remain near-stationary ($|\beta^i| \ll 1$) and out of any deep-lapse well ($\alpha \geq 0.95$ at launch). Source: `endpoint_gauge_rm.txt`.

t	α_A	$ \beta^i _A$	α_B	$ \beta^i _B$
0	1.000	0	1.000	0
4	0.960	4.5×10^{-3}	0.993	3.5×10^{-2}
8	0.946	1.7×10^{-2}	1.048	9.3×10^{-2}
12	1.179	1.2×10^{-2}	1.209	1.3×10^{-2}
15	1.172	6.9×10^{-2}	1.118	7.1×10^{-2}

sured $g_{00}(B, t)$ from t_{emit} to each arrival, the curved signal reaches B at $\tau_{\text{curved}} = 12.61$ and the matched flat-rate signal at $\tau_{\text{flat}} = 15.66$, giving

$$f_{\text{geo}}^\tau = \frac{\tau_{\text{flat}} - \tau_{\text{curved}}}{\tau_{\text{flat}}} = 19.5\%, \quad (18)$$

within 0.7 percentage points of the coordinate value 20.2%. The advantage is therefore robust against lapse/shift coordinate effects in the selected gauge: it survives conversion to the physical clock of a stationary receiver at B . This does not constitute a gauge-invariant proof—the receiver is defined by fixed coordinates, not a geometrically selected worldline—but it confirms the signal is not a coordinate artifact of the $1 + \log$ slicing.

A complementary check concerns proper *distance*. The negative-energy sources contract the spatial metric along the corridor, so the coordinate separation 14.4 overstates the proper length of the A – B line: integrating $\int \sqrt{\gamma_{xx}} dx$ in the stored RM metric stack gives 13.85 at $t = 0$, 13.60 at the stored slice nearest the headline launch ($t = 4.32$), and a value inside $[13.08, 13.67]$ throughout the flight window. Part of the coordinate advantage is therefore genuine spatial contraction sourced by the exotic matter—a physical effect, and one the ablation of Sec. VC isolates quantitatively. Even under the most conservative renormalization, comparing the curved transit $t_{\text{min}} = 11.49$ against a flat transit across the *smallest* instantaneous proper length, $(13.08 - 11.49)/13.08 = 12.2\%$, the advantage stays strictly positive. We quote the coordinate-referenced 20.2% as the headline number because Eq. (13) defines its flat baseline by the fixed coordinate endpoints, but both normalizations certify an early arrival.

F. Resolution and domain matrix

Validation comprises two complementary tests of candidate 146 (Table VII). At fixed domain $L = 128$, a resolution ladder—RC ($N = 192$), RM ($N = 256$), RF ($N = 384$)—probes continuum stability. At fixed spacing $h = 0.500$, a domain ladder—DS ($L = 96$) and DL ($L = 160$)—probes sensitivity to outer-boundary placement, all evolutions carrying non-periodic Sommerfeld

TABLE VII. Resolution and domain matrix for candidate 146. Every run uses three refinement levels above the base grid (`max_level=3`, `regrid_threshold=0.02`) and terminates at $t = 30$. Peak f_{geo} is the maximum evolving 4D null-geodesic timing advantage; frozen peak is the maximum of the frozen-slice f_{geo} time series; f_{ff} is the freely falling observer metric (Sec. IV B), taken from three matched companion integrations (`bcma*_freefall_corrected_*`) whose metric stacks are bit-identical to the runs in the other columns.

run	L	N	h	peak f_{geo}	frozen peak	f_{ff}	t_{emit}
RC	128	192	0.667	20.44%	29.15%	7.69%	4
RM	128	256	0.500	20.21%	29.45%	7.62%	4
RF	128	384	0.333	20.19%	29.54%	7.01%	4
DS	96	192	0.500	20.19%	29.21%	—	4
DL	160	320	0.500	20.22%	29.44%	—	4

(radiative) conditions on all six faces. RM is the reference configuration quoted elsewhere in this paper.

On the resolution ladder the evolving timing advantage runs $20.44\% \rightarrow 20.21\% \rightarrow 20.19\%$: monotonically decreasing, with a fine-medium relative difference of 0.13%, far inside the 10% acceptance tolerance fixed before the ladder was run. The frozen peak stays at 29.1–29.5% and the freely falling observer metric at 7.01–7.69%, the latter’s 8% fine-medium spread indicating mild but acceptable resolution dependence in the proper-time measurement. On the domain ladder, DS and DL differ from RM by at most 0.12% relative, with five of five rays reaching in every run. Outer-boundary reflections therefore do not set the reported advantage on $t \leq 30$, consistent with a causal estimate for the $L = 128$ box: in $1 + \log$ slicing, gauge modes travelling at $\sim \sqrt{2}c$ need $t \gtrsim L/(2\sqrt{2}) \approx 45$ to reach the central region.

For unequal spacings, a Richardson estimate assumes

$$Q(h) = Q_0 + Ch^p, \quad \frac{Q_c - Q_m}{Q_m - Q_f} = \frac{h_c^p - h_m^p}{h_m^p - h_f^p}. \quad (19)$$

Applied to the f_{geo} triplet at the actual spacings $(h_c, h_m, h_f) = (0.667, 0.500, 0.333)$, Eq. (19) returns an unstable exponent ($p \approx 8.7$) fixed by a fine-medium difference of one significant figure: the sequence is consistent with rapid convergence but cannot pin down p . The anomaly is expected here, because f_{geo} is not a pointwise field value but an arrival-time functional—root-found from a detector crossing along five rays integrated through a stack stored at finite cadence on a moving refinement hierarchy, then reported as the best member of the fan over a discrete t_{emit} sweep. Regridding times, stack interpolation, and a different ray or emission time winning at a different resolution all make the dependence non-smooth rather than Ch^p , and once the smooth truncation error has fallen to the 10^{-3} level seen here they dominate the increments the estimator is built from. We therefore measure the observed order on the constraint norms instead (Fig. 10a): over the ignition window $1.5 \leq t \leq 5$ the Hamiltonian residual converges

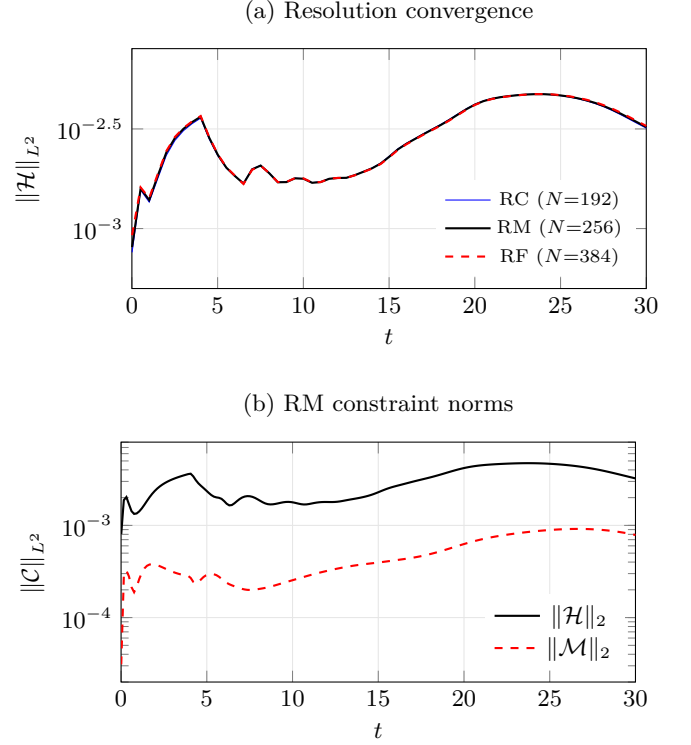


FIG. 10. Constraint diagnostics. (a) Hamiltonian constraint norm across the Richardson ladder; the three resolutions overlap closely with observed order $p_{\mathcal{H}} \approx 3.3$ (median) in the ignition window. (b) Constraint norms in the headline RM run; the complete 3000-step diagnostic is plotted from a decimated table; maxima are evaluated from the full data.

at median order $p_{\mathcal{H}} \approx 3.3$ and the momentum residual at $p_{\mathcal{M}} \approx 2.8$, consistent with the high-order finite-difference scheme. We accept the 20.2% shortcut as continuum-stable without quoting a formal extrapolated value.

G. Evolution constraints

For RM, the volume-weighted norms remain below $\|\mathcal{H}\|_2 = 4.73 \times 10^{-3}$ and $\|\mathcal{M}\|_2 = 9.16 \times 10^{-4}$ through $t = 30$ (Fig. 10b). These are comparable to the search-tier values and remain bounded throughout the evolution. The constraint norms peak during the early driven phase ($t \lesssim 4$) and then plateau or slowly grow during the free evolution, consistent with the dispersing matter producing less active refinement at late times.

H. Evolution with the PD controller disabled

Because the PD trajectory controller (Sec. IIF) is an external, non-Lagrangian source, we repeat the HQ evolution of candidate 146 with the controller disabled from $t = 0$ ($t_{\text{pump}} = 0$), holding all other settings fixed ($L = 128$, $N = 256$, three refinement levels, $t = 30$). Ta-

TABLE VIII. Matched HQ comparison of candidate 146 with the PD controller on through $t = 4$ (RM) and disabled from $t = 0$ (PF-RM). Both runs use $L = 128$, $N = 256$, and $t = 30$. Evolving f_{geo} is the 4D null-geodesic timing advantage; E_{pump} is the integrated pump work recorded in `collapse_diagnostics.dat`; S is the final `general_ftl` score.

metric	RM ($t_{\text{pump}} = 4$)	PF-RM ($t_{\text{pump}} = 0$)
evolving f_{geo}	20.21%	20.46%
rays / h quality	5/5, pass	5/5, pass
best t_{emit}	4	12
E_{pump}	1.6×10^{-17}	0
confined ($t=0 \rightarrow 30$)	72% $\rightarrow$ 11%	72% $\rightarrow$ 2%
rms spread ratio	3.27	3.07
final score S	126.3	47.0

ble VIII compares this pump-free integration (PF-RM) with the default pump-on twin (RM; $t_{\text{pump}} = 4$).

With the PD term off, the evolving timing advantage remains $f_{\text{geo}} = 20.46\%$ —within 1.2% relative of the pump-on value—with all five rays reaching and the null-Hamiltonian quality gate satisfied. The integrated pump work vanishes identically, so the advantage cannot be ascribed to external PD energy injection. The channel does open more slowly: the emission optimum shifts from $t_{\text{emit}} = 4$ to $t_{\text{emit}} = 12$ and migrates from the x axis to the z axis of the probe, while the peak amplitude is essentially unchanged. Because $t_{\text{emit}} = 12$ is the last point of the emission sweep, the pump-free value is strictly a lower bound on that run’s peak advantage; the channel may still be opening when the sweep window closes.

The controller’s principal effect is therefore on matter retention, not on f_{geo} : at $t = 30$ the pump-on run still confines $\sim 11\%$ of the matter against $\sim 2\%$ without it. That is also why the composite score falls from $S = 126.3$ to $S = 47.0$ —confinement-gated terms penalize the faster dispersal, while the timing advantage itself is unchanged. In both integrations the shortcut remains a transient precursor coexisting with dispersing exotic matter.

We nonetheless keep RM, not PF-RM, as the headline configuration. The first reason is consistency with the search: the entire MAP-Elites and CMA-ES campaign ran with the controller enabled through $t_{\text{pump}} = 4$, so candidate 146 is an elite *of that protocol*. PF-RM re-evaluates the same genome under a different one; it is a control, not an independently optimized solution, and its emission optimum has already migrated to the last point of the sweep window. The second reason is the five-fold confinement difference above. Structural persistence is what distinguishes a precursor from a burst of dispersing scalar field, and it is the property the controller buys. Read as engineering rather than as a Lagrangian, the PD term is a placeholder for the station-keeping actuation any real device would require, and the configuration worth reporting is the one under active control. Nothing

in the transport certificate turns on the choice: the headline ray launches after the controller is off, and the two protocols agree on f_{geo} to within 1.2% relative. PF-RM’s role is to establish that the control is not manufacturing the geometry.

VI. DISCUSSION

A. The dispersion problem

Candidate 146 should be read against the many configurations that did not work. Across roughly a dozen ansatz families and several full MAP-Elites campaigns we repeatedly found the same failure mode: the supporting matter disperses and the transport channel closes with it. Spherical-harmonic shells yielded a single weak hit in ~ 200 evaluations. Free massive scalar packets have no bound state and blow apart almost immediately. Bicomplex phantom lumps *without* Q-ball self-binding (an earlier, pre-soliton family) opened a gauge-invariant shortcut but lost $\sim 98\%$ of their structure by $t = 16$. Boosted and stiffened Q-ball variants improved early confinement but still sheared apart once placed on an orbit. CMA-ES bought back some persistence (Sec. IIIB), but candidate 146 still ends the production window at $\sim 11\%$ confinement.

Beyond dispersal, the HQ runs develop a corroborated trapped surface at late times ($t \approx 27$ in the resolution ladder, $t \approx 24$ in the pump-free twin), with a minimum expansion scalar $\theta_+ \approx -0.9$ and $r_{\text{AH}} \approx 10.8$ code units. The headline geodesic launches at $t_{\text{emit}} = 4$ and arrives at $t \approx 15.5$, well before any apparent horizon forms, so the transport certificate is unaffected. The collapse is itself a symptom of transience: once the exotic matter disperses, the residual positive-energy content exceeds the threshold for gravitational collapse. A transient shortcut is reproducible and robust here; a long-lived, well-confined channel remains an open problem.

B. Canonical-only (positive-energy) search bound

Reported positive-energy warp solitons [14, 15] and their critique [11] make the positive-energy sector the natural next target for an automated search. To test whether the same five-lump bicomplex Q-ball genome can produce a gauge-invariant timing advantage without phantom sources, we reran the MAP-Elites campaign of Sec. III with every per-lump exotic amplitude pinned to zero, retaining the canonical Klein–Gordon sector, retrograde-orbit bias, and post-load constraint gates. Over 200 evaluations (105 GPU-accepted; 95 rejected at the post-load Hamiltonian L^2 gate), the archive champion reached only $S \approx 47.4$ (eval. 130), and the evolving geodesic fraction satisfied

$$\max_{\theta \in \mathcal{A}_{\text{can}}} f_{\text{geo}}^{\text{evol}}(\theta) = 0 \quad (N_{\text{ok}} = 105). \quad (20)$$

Several accepted genomes developed superluminal local coordinate signal speeds ($\max v_{\text{loc}} > 1$, a gauge-dependent diagnostic) or early apparent horizons, but none completed a null-timing advantage under the evolving-geodesic certificate. Within this ansatz and search budget, a completed $f_{\text{geo}} > 0$ therefore appears to require the phantom sector used by candidate 146; the positive-energy search remains open for broader matter families and longer t_{stop} .

C. Limitations of the matter-first approach

The matter-first approach (Sec. I) is complementary to geometry-first [5]: every candidate is constraint-satisfying by construction, but the search is confined to the chosen matter model. Three specific limitations follow.

1. **Ansatz dependence.** The pipeline discovers only what the parameterized matter family can express. The 39-dimensional bicomplex Q-ball genome covers orbital arrangements of five solitons but cannot represent, e.g., anisotropic fluids, higher-spin fields, or engineered equation-of-state shells. Broadening the matter sector is the most direct route to new mechanism families.
2. **$\mathbb{R}^3$ spatial topology.** Both the constraint solve and the Cauchy evolution operate on a topologically trivial slice ($\mathbb{R}^3$ with Sommerfeld radiative boundaries). Features that exist only in the full four-dimensional manifold—wormhole throats, handles, other nontrivial $\mathbb{R}^4$ structures—cannot be represented on the initial slice and are invisible to the search. Reaching them would need an extended initial-data formalism (puncture or excision methods) or a geometry-first seed that is subsequently evolved.
3. **Ignition-phase controller.** The PD pump (Sec. IIF) is a non-Lagrangian external source; results without it (Sec. V H) are comparable. A fully self-consistent confinement mechanism would remove the need for any external drive.

D. Geometry-first kinematic ceiling

The canonical-only bound of Sec. VIB asks how far the search reaches within one matter model. A complementary question is purely kinematic: setting sourceability aside, how large a frozen-slice shortcut is available to *any* asymptotically flat stationary metric of this scale? To answer it we ran a deliberately geometry-first companion search—the very regime the Introduction rejects for dynamics—in order to price what the matter-first pipeline leaves on the table.

TABLE IX. Frozen geometry-first kinematic ceiling from the stationary MAP-Elites atlas (grid $N = 64$, $L = 32$; localized null-geodesic probe; no evolution). f_{shift} is the morphology descriptor and $E_- = -\int_{\rho < 0} \rho dV$ the exotic cost. Frozen f_{geo} is geometry-first and *not* comparable to the evolving certificate of Table V. The tunnel/lens/throat rows are single-parameter zero-shift screens ($N = 48$); the last row is the best geometry admitted under a hard $E_- \leq 0.1$ ban.

family	f_{shift}	frozen f_{geo}	E_-
shift tube (Alcubierre)	≈ 0.99	48.5%	3–7
curvature-dominated elite	≤ 0.05	21.8%	~ 4
compression tunnel (screen)	0.00	9.4%	—
conformal lens (screen)	0.00	5.5%	—
wormhole throat (screen)	0.00	3.5%	—
low-exotic ban ($E_- \leq 0.1$)	—	6.8%	≤ 0.1

Here a stationary metric is painted directly from a genome: compact-support radial-basis deformations of the lapse, shift, and symmetric log-spatial-metric S (with $\gamma = \exp S$ guaranteeing positive-definiteness and asymptotic flatness), plus an analytic tail of coherent topologies—an Alcubierre shift tube, an anisotropic “compression tunnel,” a conformal “lens” well, and a Morris–Thorne–flavored “throat.” No initial-value problem is solved: the implied stress-energy is read off as $T_{\mu\nu} = G_{\mu\nu}/8\pi$, and only the *frozen* single-slice f_{geo} is scored, so these numbers are *not* comparable to Table V. MAP-Elites [12] tiles the archive by morphology (shift fraction f_{shift} , running from 0 for pure spatial curvature to 1 for a pure shift tube) against exotic cost $E_- = -\int_{\rho < 0} \rho dV$. Three warm-started campaigns totaling ~ 2000 evaluations populate it, the last hard-rejecting $E_- > 0.1$. Two settings are screening-tier rather than continuum: the coherent analytic topologies had to be added because free-form Gaussian bumps collapse into a maximal-shift monoculture, and the frozen probe reads correctly only with endpoints localized about the genome support (~ 8 percentage points against diluted full-box endpoints) and $h \leq 0.5$ (at $h = 2$ the shift wall is unresolved and under-reads by nearly a factor of five).

Two features of the atlas (Table IX) bear on the matter-first result. First, strength is monopolized by the shift channel: the frozen ceiling reaches 48.5% for near-pure shift tubes ($f_{\text{shift}} \approx 0.99$), and of the 781 accepted geometries with $f_{\text{geo}} > 10\%$, 67% sit at $f_{\text{shift}} \geq 0.5$ against only 4.5% at $f_{\text{shift}} \leq 0.05$. Zero-shift curvature topologies do open genuine shortcuts—the best curvature-dominated elite reaches 21.8%—but they are systematically weaker. Second, none of these shortcuts is cheap: every geometry with a positive frozen shortcut carries substantial negative energy ($E_- \sim 3\text{--}7$), and hard-rejecting $E_- > 0.1$ leaves a best admissible shortcut of 6.8%, a factor ~ 7 below the unconstrained ceiling. Shift-tilted and curvature-contracted shortcuts alike demand NEC violation, in line with the classical theorems cited in Sec. I.

Placed against the atlas, candidate 146 stops looking like a weak imitation of an Alcubierre bubble and starts looking like a saturated optimum on the other branch. Its transport is carried by spatial contraction and lapse elevation with the shift a net drag (Sec. VC), so its morphology belongs at the low- f_{shift} end of the archive, where the ceiling is 21.8% rather than 48.5%. Delivering 20.2% under full Cauchy evolution, the matter-first optimizer came within 1.6 percentage points of the frozen ceiling of the branch it can actually source—while leaving the stronger branch untouched.

That split is the substance of the sourceability barrier. The shift tube needs a coherent, one-directional momentum density S_i concentrated in a thin wall; a smooth superposition of orbiting Q-balls sources a bipolar S_i instead (Fig. 8), and what confinement it achieves decays (Sec. VIA). The conformal well needs only a localized negative energy density, which a phantom lump supplies directly. Kinematic strength and dynamical sourceability are therefore traded against each other, and not smoothly: the branch that is easy to source is capped at roughly half the frozen ceiling of the branch that is not. Promoting an atlas shift-tube motif through GR_{Tresna} and GR_{Teclyn} is the obvious next experiment.

The atlas remains a bound rather than a claim. Scored on a single stationary slice with matter merely inferred, a positive frozen f_{geo} need not survive a Cauchy evolution—exactly the false-positive mode of the negative control in Sec. IV—free-fall reception is not computed, and the search is confined to its own ansatz family.

E. Laboratory Casimir bounds on null-timing shortcuts

The exotic negative energy supporting candidate 146 is a continuum phantom sector, not a laboratory vacuum stress. For ideal parallel plates of separation a , the Casimir energy density [24] is

$$\rho_{\text{cas}} = -\frac{\pi^2 \hbar c}{720 a^4}. \quad (21)$$

Under linearized semiclassical gravity, even an optimistic $a = 1\text{ nm}$ gap in a 1 km waveguide yields $f_{\text{geo}}^{\text{bound}} \sim 10^{-50}$ —nearly 50 orders of magnitude below candidate 146’s 20.2%, and ~ 37 orders below the best atomic-clock resolution, consistent with quantum-inequality bounds [25]. Proposed amplification routes (dynamical Casimir effect [26], plasmonic metamaterials, slow-light media) remain open modelling targets. The automated search discovers large f_{geo} only because it explores continuum exotic matter far outside laboratory stress-energy budgets.

F. Computational cost

All runs used a single 8-GPU node (NVIDIA H100 80 GB HBM3), with MAP-Elites and CMA-ES dispatching one GR_{Teclyn} evolution per GPU and concurrent GR_{Tresna} solves on CPU (MPI, 8 ranks per solve). A 200-evaluation MAP-Elites campaign at search resolution takes about two wall-clock hours and the 150-evaluation CMA-ES refinement a similar time. Production validation is sequential: RM fits on one GPU, while the fine Richardson run RF ($N = 384$) needed three for memory. Total wall-clock time from the start of MAP-Elites to the completed resolution matrix was approximately ten hours.

G. Future applications of the pipeline

Within the present target, the immediate priorities are longer t_{stop} integrations to track late-time dispersal, positive-energy matter families beyond the canonical-only bound of Sec. VIB, non-dispersing matter models, and confinement world-tube objectives.

The architecture, however, is not specific to null-timing shortcuts. Any question of the form “which initial matter configuration optimizes geometric observable $\mathcal{O}$?” can be attacked by replacing the scoring function while reusing the genome, constraint solver, and evolution infrastructure.

Directional gravitational-wave generation. A parallel campaign (in preparation) replaces f_{geo} with a beaming-weighted Weyl-scalar power score. Early runs produce $\sim 30\%$ directional beaming gain with mean Weyl power $P \sim 10^{-6}$ – 10^{-4} , showing the optimizer can steer toward anisotropic radiation from matter that is individually spherically symmetric.

Other candidate targets. (i) Gravitational-lensing amplification—optimizing a matter shell to maximize the magnification of a background source at a specified image plane; (ii) critical-collapse threshold mapping—locating the boundary between dispersal and black-hole formation in multi-lump parameter space; (iii) boson-star merger waveform exploration—seeding binary or multi-body initial data and scoring the resulting gravitational-wave energy or kick velocity. In each case the pipeline inherits the hard gates (Hamiltonian constraint, collapse diagnostics) that reject numerical artefacts, addressing the reward-hacking failure modes common to optimization in nonlinear PDE solvers.

VII. CONCLUSION

We presented an automated matter-first discovery pipeline in which MAP-Elites proposes scalar configurations, CMA-ES refines the archive champion, and 4D null geodesics through the evolved spacetime test whether light arrives earlier than in flat space. Applied to a

bicomplex five-lump Q-ball genome, it discovered candidate 146, a reproducible, continuum-stable numerical warp precursor whose freely falling observers receive light 7.6% early by their own clocks ($f_{\text{ff}} = 7.0\text{--}7.7\%$ across the ladder; $f_{\text{geo}} = 20.2\%$ between fixed coordinate endpoints; Table VII), and which survives with the ignition controller disabled (Sec. V H). The shortcut is transient—the matter disperses and the residual geometry collapses to an apparent horizon, so this is not a functional warp drive—and, within this ansatz, requires the phantom sector: a matched canonical-only search found no $f_{\text{geo}} > 0$ (Sec. VIB), while semiclassical Casimir bounds (Sec. VIE) place laboratory vacuum stresses ~ 50 orders of magnitude short of the required exotic budget.

What the pipeline changes is not the size of the number but the question it answers. Warp geometries have been drawn on paper for three decades without anyone checking whether they can be sourced by a valid initial-value problem and survive their own evolution. When that check is enforced, the optimum moves. The strongest kinematic object in the landscape is still the Alcubierre shift tube at 48.5%, and our optimizer never went near it; what it found instead was a conformal-contraction precursor on the weaker curvature branch, which it drove to within 1.6 percentage points of that branch’s ceiling (Sec. VID). Frame dragging, the mechanism the warp-drive literature is built on, turns out to work *against* the shortcut here (Sec. VC). This sourceability barrier—not the energy conditions, which both branches violate—is what selects the shape of a dynamically viable warp precursor, and mapping it is the agenda this falsification stack opens up. The same stack transfers unchanged to the positive-energy sector and to other geometric observables (Sec. VIG).

CODE AND DATA AVAILABILITY

The code that generates every result in this paper—the matter-first pipeline, constraint solver and evolution drivers, and the analysis scripts behind every figure and table—is publicly available on the `develop` branch of the `GRTEclyn` and `GRTesna` repositories (<https://github.com/Nikchik-coder/GRTEclyn/tree/develop> and <https://github.com/Nikchik-coder/GRTesna/tree/develop>). A lightweight audit pack that mirrors every plotted number and HQ claim—evolving-geodesic reports, constraint histories, and confinement timeseries for the RC/RM/RF resolution ladder, the DS/DL domain ladder, and the pump-free PF-RM twin, together with candidate-146 provenance and the canonical-only MAP-Elites archive (Sec. VIB)—lives on that branch under `results/matter-first-automated-discovery-of-transient-spacetime-shortcuts/`. Figure files and the reduced numerical tables used by TikZ/PGFPlots are under `research/neuralspacetime/`, and the Casimir bound script (Sec. VIE) is `research/neuralspacetime/casimir_null_timing_bounds.py`.

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